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21 December 1987
SUPERSEDING
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1 December 1976

MILITARY SPECIFICATION

*CLOTH, INTERLINING, COTTON OR SYNTHETIC WARP, AND SPUN HAIR-WOOL OR RAYON FILLING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers interlining cloth constructed from cotton or synthetic warp yarns and a blend of spun goat-hair and wool or spun goat-hair and rayon filling yarn.

1.2 Classification. The cloth shall be of the following types and classes as specified (see 6.2):

Type I	- Medium weight
Type II	- Heavy weight
Type III	- Light weight
Class 1	- Natural
Class 2	- Dyed

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Rules and Regulations Under the Wool Products Labeling Act (16 CFR Part 300)

(Application for copies should be addressed to the Federal Trade Commission, Correspondence Branch, Pennsylvania at Sixth Street, N.W., Washington, DC 20580 or to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

Rules and Regulations Under the Textile Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

* 3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and finish appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material. (See 6.4.)

* 3.2.1 Cotton or synthetic (warp). The warp yarn shall be made from 100 percent cotton, 100 percent polyester, any combination of polyester and cotton, or from 45 to 100 percent polyester with the remaining percentage rayon.

3.2.2 Goat-hair and wool (filling). The goat-hair and wool shall be blended together and then spun on the worsted system into a singles yarn. The wool shall be of staple fleece or pulled wool.

3.2.2.1 Animal fiber content (hair-wool filling). The cloth shall analyze no less than 50 percent animal fiber based on the dry desized weight when tested as specified in 4.2.3.

3.2.3 Goat-hair and rayon (filling). The goat-hair and rayon shall be blended equally together and spun into singles yarn.

* 3.2.3.1 Animal fiber content (hair-rayon filling). Type I and type III cloth shall analyze no less than 25 percent and type II cloth shall analyze no less than 22 percent animal fiber based on the dry weight when tested as specified in 4.2.3.

3.3 Color. The color of the class 2 cloth shall be Black 341 (see 6.5) and shall match the standard sample (see 6.3).

3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.3.

* 3.3.2 Matching. The color of the dyed cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

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3.3.3 Colorfastness. The dyed and finished cloth shall show fastness to wet drycleaning, perspiration, and steam pressing equal to or better than the standard sample or equal to or better than a rating of "good" when tested as specified in 4.2.3.

* 3.4 Physical requirements. The physical requirements of the finished cloth shall be as specified in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

Type	Weight, ounces per sq yd		Yarn per inch, minimum		Breaking strength (pounds), minimum		Stiffness 1/ (load-pounds)
	Min	Max	Warp	Filling	Warp	Filling	Min
I	7.0	8.0	42	36	50	45	0.011
II	7.5	9.0	42	36	50	45	0.016
III	5.0	7.0	42	36	50	45	0.005

1/ Measured parallel with filling direction.

3.4.1 Weave. The weave shall be plain.

* 3.4.2 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringes.

3.5 Finish. The class 1 cloth shall have an unbleached natural finish. The class 2 cloth shall be dyed.

3.5.1 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 8.0 percent when tested as specified in 4.2.3.

3.5.2 pH. The pH value of the water extract of the dyed and finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.

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* 3.6 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 3.0 percent in the warp direction nor more than 2.0 percent in the filling direction when tested as specified in 4.2.3. The preshrinkage processes used shall not be identified by name or trademark on the cloth, ticket, or package.

3.7 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 40 yards. Each length shall be put-up in full width rolls as specified in 5.1.

3.8 Wool content label. The wool cloth manufactured under this document shall be labeled on the ticket in accordance with the Wool Products Labeling Act.

3.9 Workmanship. The finished cloth shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsiblity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The lot shall be expressed in units of linear yards, full width, of the finished cloth. The sample unit shall be 1 yard. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units (yards), shall be 6.5. The number of rolls from which the sample yardage is to be selected shall be in accordance with table II. The sample yardage shall be apportioned equally among the selected rolls.

Defects

Hole, cut, or tear
Smash, multiple floats, spots or stains
Embedded crease or wrinkle
Broken or missing pick or end

4.2.2.2 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The size (number of rolls selected as sample) and the acceptance number shall be as shown in table II.

Defects

Width less than specified 1/
Overall uncleanness 1/
Baggy, ridgy, or wavy cloth

1/ Shall not be scored unless it occurs for more than 5 yards per roll.

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TABLE II. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1200 1/	3	0
1,201 up to and including 3,200	5	0
3,201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	1
35,001 up to and including 150,000	20	2
150,001 and over	32	3

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.2.2.3 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.2.4 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and finish appearance. Any roll in the sample off shade or shaded side to side, side to center, or end to end, or any roll that does not have the same appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

* 4.2.3 End item testing. The cloth shall be tested for the characteristics listed in table III. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards, full width, of the finished cloth. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be

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expressed in units of yards. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Warp yarn:		
Cotton identification	3.2.1	1200 1/
Synthetic fiber identification	3.2.1	1600 1/
Filling yarn:		
Spun goat-hair and wool identification	3.2.2	1/
Spun goat-hair and rayon identification	3.2.3	1/
Yarn (singles)	3.2.2 and 3.2.3	1/
Animal fiber content	3.2.2.1 and 3.2.3.1	2100
Presence of labile sulfur	3.3.1	2020
Colorfastness to:		
Wet drycleaning	3.3.3	5622
Perspiration	3.3.3	5680
Steam pressing	3.3.3	5640
Weight	3.4	5041
Yarns per inch	3.4	5050
Breaking strength	3.4	5100
Stiffness (filling only)	3.4	5202
Weave	3.4.1	Visual 2/

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TABLE III. End item tests - Continued

Characteristic	Requirement paragraph	Test method
Nonfibrous materials	3.5.1	2611
pH	3.5.2	2811
Shrinkage	3.6	5558 3/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and the results shall be reported as "pass" or "fail".

3/ Method 5558 shall be used except that the specimens shall be dried as specified in Method 5550. The cloth shall be kept submerged for a period of 30 minutes in water containing a minimum amount of a suitable wetting agent to thoroughly wet out the fabric.

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

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6. NOTES

6.1 Intended use. The cloth is intended for use as an interlining in clothing items.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and class required (see 1.2).
c. Width of cloth required (see 3.4.2).
d. Length required when other than specified (see 3.7).
e. Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.3 Standard sample. For access to samples, address the contracting activity issuing the invitation bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.5 Dyestuff formulation, Black 341. A suggested but not mandatory formulation for Black 341 is as follows:

Direct Black RL
Neutral Premetallized Black C-BL
Aftertreated with copper sulfate

* 6.6 Subject term (key word) listing.

Cloth
Clothing
Interlining

6.7 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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